PROJECT PROPOSAL – MATH 563

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Algorithm Allocation

We will for the most part work on all four deblurring algorithms collaboratively. As the deadline approaches, we will then allocate responsibilities to ensure that each algorithm is well done. Initial allocation, subject to change, will be as follows:

1. Primal Douglas-Rachford Splitting (Spingarn's Method) (Alexandre St-Aubin)
2. Primal-Dual Douglas Rachford Splitting (Jonathan Campagna)
3. ADMM (Dual Douglas-Rachford) (Edmund Wu)
4. Chambolle-Pock Method (Elliot Forcier-Poirier)

Matlab Implementation

Matlab will be used for coding the deblurring algorithms, making use of its built-in functions for matrix operations, optimization routines, and powerful visualization tools.

GitHub Repository Collaboration

We plan to use GitHub for collaborative coding and version control, creating a private repository exclusive to team members. This will facilitate an organized and overview of our collective work

Final Report – Overleaf Collaboration

The final report will be prepared using LaTeX, per the instructions. We will collaborate on Overleaf, allowing for simultaneous writing. Specific sections will be assigned to each team member, as in the coding portion.